

HTML Tables and Forms Complete Reference

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HTML Tables

Basic Table Structure

```
<table>
  <tr>                                <!-- Table row -->
    <td>Cell 1</td>                  <!-- Table data/cell -->
    <td>Cell 2</td>
    <td>Cell 3</td>
  </tr>
  <tr>
    <td>Cell 4</td>
    <td>Cell 5</td>
    <td>Cell 6</td>
  </tr>
</table>
```

Complete Table with Headers

```
<table>
  <tr>                                <!-- Table header -->
    <th>Name</th>
    <th>Age</th>
    <th>City</th>
  </tr>
  <tr>
    <td>John Doe</td>
    <td>30</td>
    <td>New York</td>
  </tr>
  <tr>
    <td>Jane Smith</td>
    <td>25</td>
    <td>Los Angeles</td>
  </tr>
```

```
</tr>
</table>
```

Table Structure Elements

Semantic Table Structure

```
<table>
  <caption>Employee Information Table</caption>      <!-- Table
caption/title -->

  <thead>                                              <!-- Table header
section -->
    <tr>
      <th scope="col">Employee ID</th>              <!-- Column header
-->
      <th scope="col">Name</th>
      <th scope="col">Department</th>
      <th scope="col">Salary</th>
    </tr>
  </thead>

  <tbody>                                              <!-- Table body
section -->
    <tr>
      <td>001</td>
      <td>John Doe</td>
      <td>Engineering</td>
      <td>$75,000</td>
    </tr>
    <tr>
      <td>002</td>
      <td>Jane Smith</td>
      <td>Marketing</td>
      <td>$65,000</td>
    </tr>
    <tr>
      <td>003</td>
      <td>Bob Johnson</td>
      <td>Sales</td>
      <td>$55,000</td>
    </tr>
  </tbody>

  <tfoot>                                              <!-- Table footer
section -->
    <tr>
      <td colspan="3">Total Employees</td>          <!-- Spans 3
columns -->
      <td>3</td>
    </tr>
  </tfoot>
</table>
```

Table Structure Elements Explained

- `<table>` - Container for entire table
 - `<caption>` - Table title/description (optional)
 - `<thead>` - Groups header content
 - `<tbody>` - Groups body content (main data)
 - `<tfoot>` - Groups footer content
 - `<tr>` - Table row
 - `<th>` - Table header cell
 - `<td>` - Table data cell
-

Table Styling and Attributes

Border Attributes

```
<!-- Basic border -->
<table border="1">
  <tr>
    <td>Cell with border</td>
    <td>Another cell</td>
  </tr>
</table>

<!-- Border with custom width -->
<table border="2">
  <tr>
    <td>Thicker border</td>
    <td>Same border</td>
  </tr>
</table>

<!-- No border -->
<table border="0">
  <tr>
    <td>No border</td>
    <td>Clean look</td>
  </tr>
</table>
```

Spacing Attributes

```
<!-- Cellpadding - space inside cells -->
<table border="1" cellpadding="10">
  <tr>
    <td>More space inside</td>
    <td>Comfortable reading</td>
  </tr>
</table>

<!-- Cellspacing - space between cells -->
<table border="1" cellspacing="5">
  <tr>
    <td>Space between cells</td>
    <td>Separated cells</td>
  </tr>
</table>
```

```

<!-- Combined spacing -->
<table border="1" cellpadding="8" cellspacing="3">
  <tr>
    <td>Both spacings applied</td>
    <td>Well-spaced table</td>
  </tr>
</table>

```

Width and Height Control

```

<table border="1" width="100%">                                <!-- Full width --
>
  <tr>
    <td width="30%">30% width</td>                             <!-- Column width
-->
    <td width="70%">70% width</td>
  </tr>
</table>

<table border="1" width="500">                                <!-- Fixed pixel
width -->
  <tr>
    <td height="50">Fixed height cell</td>                     <!-- Row height --
>
    <td>Normal height</td>
  </tr>
</table>

```

Alignment Attributes

```

<table border="1" align="center">                                <!-- Table
alignment -->
  <tr>
    <td align="left">Left aligned</td>                          <!-- Horizontal
alignment -->
    <td align="center">Center aligned</td>
    <td align="right">Right aligned</td>
  </tr>
  <tr>
    <td valign="top" height="60">Top</td>                       <!-- Vertical
alignment -->
    <td valign="middle">Middle</td>
    <td valign="bottom">Bottom</td>
  </tr>
</table>

```

Background Colors and Images

```

<table border="1">
  <tr bgcolor="#f0f0f0">                                         <!-- Row
background -->
    <td>Gray row</td>
    <td bgcolor="#ffcccc">Pink cell</td>                         <!-- Cell
background -->
  </tr>
  <tr>
    <td background="pattern.jpg">Background image</td> <!--
Background image -->

```

```
        <td>Normal cell</td>
    </tr>
</table>
```

Advanced Table Features

Cell Spanning

```
<table border="1">
  <!-- Column spanning -->
  <tr>
    <td colspan="3">This cell spans 3 columns</td> <!-- Horizontal
span -->
  </tr>
  <tr>
    <td>Column 1</td>
    <td>Column 2</td>
    <td>Column 3</td>
  </tr>

  <!-- Row spanning -->
  <tr>
    <td rowspan="2">Spans 2 rows</td> <!-- Vertical
span -->
    <td>Regular cell</td>
    <td>Regular cell</td>
  </tr>
  <tr>
    <!-- First cell continues from above -->
    <td>Bottom middle</td>
    <td>Bottom right</td>
  </tr>

  <!-- Complex spanning -->
  <tr>
    <td rowspan="2" colspan="2">Spans 2 rows and 2 columns</td>
    <td>Right cell</td>
  </tr>
  <tr>
    <td>Bottom right</td>
  </tr>
</table>
```

Accessible Tables

```
<table border="1">
  <caption>Sales Data by Quarter</caption>

  <thead>
    <tr>
      <th scope="col" id="product">Product</th> <!-- Column
scope -->
      <th scope="col" id="q1">Q1</th>
      <th scope="col" id="q2">Q2</th>
      <th scope="col" id="total">Total</th>
    </tr>
```

```

        </thead>

        <tbody>
        <tr>
            <th scope="row" id="laptops">Laptops</th>      <!-- Row scope -
->
            <td headers="laptops q1">150</td>              <!-- Headers
reference -->
            <td headers="laptops q2">200</td>
            <td headers="laptops total">350</td>
        </tr>
        <tr>
            <th scope="row" id="tablets">Tablets</th>
            <td headers="tablets q1">100</td>
            <td headers="tablets q2">120</td>
            <td headers="tablets total">220</td>
        </tr>
        </tbody>
    </table>

```

Nested Tables

```

<table border="1">
    <tr>
        <td>Regular cell</td>
        <td>
            <!-- Nested table -->
            <table border="1" width="100%">
                <tr>
                    <td>Nested cell 1</td>
                    <td>Nested cell 2</td>
                </tr>
                <tr>
                    <td>Nested cell 3</td>
                    <td>Nested cell 4</td>
                </tr>
            </table>
        </td>
    </tr>
</table>

```

HTML Forms

Basic Form Structure

```

<form action="process.php" method="post" target="_blank">
    <!-- Form elements go here -->
    <input type="submit" value="Submit">
</form>

```

Form Attributes

- action="url" - Where to send form data
- method="post" or method="get" - How to send data
- target="_blank" - Where to open response

- enctype="multipart/form-data" - For file uploads
- autocomplete="on" or autocomplete="off" - Browser autocomplete
- novalidate - Disable HTML5 validation

Complete Form Example

```

    <form action="submit.php" method="post" enctype="multipart/form-
data">
    <fieldset>
        <legend>Personal Information</legend>

        <label for="firstName">First Name:</label>
        <input type="text" id="firstName" name="firstName" required>

        <label for="lastName">Last Name:</label>
        <input type="text" id="lastName" name="lastName" required>

        <label for="email">Email:</label>
        <input type="email" id="email" name="email" required>
    </fieldset>

    <fieldset>
        <legend>Preferences</legend>

        <label for="newsletter">
            <input type="checkbox" id="newsletter" name="newsletter"
value="yes">
            Subscribe to newsletter
        </label>
    </fieldset>

    <input type="submit" value="Submit Form">
    <input type="reset" value="Clear Form">
</form>

```

Form Input Types

HTML4 Input Types

```

<!-- Text input -->
<label for="username">Username:</label>
<input type="text" id="username" name="username" size="20"
maxLength="50">

<!-- Password input -->
<label for="password">Password:</label>
<input type="password" id="password" name="password" size="20">

<!-- Hidden input -->
<input type="hidden" name="userId" value="12345">

<!-- Checkbox -->
<input type="checkbox" id="agree" name="terms" value="agreed">
<label for="agree">I agree to terms and conditions</label>

<!-- Radio buttons -->

```

```

<input type="radio" id="male" name="gender" value="male">
<label for="male">Male</label>

<input type="radio" id="female" name="gender" value="female">
<label for="female">Female</label>

<input type="radio" id="other" name="gender" value="other">
<label for="other">Other</label>

<!-- Submit and Reset buttons -->
<input type="submit" value="Send Form">
<input type="reset" value="Clear Form">

<!-- Generic button -->
<input type="button" value="Click Me" onclick="alert('Clicked!')">

<!-- File upload -->
<label for="upload">Choose file:</label>
<input type="file" id="upload" name="document"
accept=".pdf,.doc,.docx">

```

HTML5 Input Types

```

<!-- Email validation -->
<label for="email">Email Address:</label>
<input type="email" id="email" name="email"
placeholder="user@example.com">

<!-- URL validation -->
<label for="website">Website:</label>
<input type="url" id="website" name="website"
placeholder="https://example.com">

<!-- Telephone -->
<label for="phone">Phone Number:</label>
<input type="tel" id="phone" name="phone" placeholder="(123) 456-
7890">

<!-- Number input -->
<label for="age">Age:</label>
<input type="number" id="age" name="age" min="18" max="100" step="1"
value="25">

<!-- Range slider -->
<label for="volume">Volume:</label>
<input type="range" id="volume" name="volume" min="0" max="100"
step="10" value="50">

<!-- Date inputs -->
<label for="birthday">Birthday:</label>
<input type="date" id="birthday" name="birthday" min="1900-01-01"
max="2024-12-31">

<label for="appointment">Appointment Time:</label>
<input type="time" id="appointment" name="appointment">

<label for="meeting">Meeting DateTime:</label>
<input type="datetime-local" id="meeting" name="meeting">

<label for="month">Expiry Month:</label>

```



```

<input type="month" id="month" name="expiry">

<label for="week">Vacation Week:</label>
<input type="week" id="week" name="vacation">

<!-- Color picker -->
<label for="theme">Theme Color:</label>
<input type="color" id="theme" name="themeColor" value="#ff0000">

<!-- Search -->
<label for="search">Search:</label>
<input type="search" id="search" name="query" placeholder="Search
our site...">

<!-- Multiple file upload -->
<label for="photos">Upload Photos:</label>
<input type="file" id="photos" name="photos" multiple
accept="image/*">

```

Other Form Elements

```

<!-- Textarea -->
<label for="comments">Comments:</label>
<textarea id="comments" name="comments" rows="5" cols="40"
placeholder="Enter your comments here..." wrap="soft">
</textarea>

<!-- Select dropdown -->
<label for="country">Country:</label>
<select id="country" name="country">
  <option value="">Select a country</option>
  <option value="us">United States</option>
  <option value="ca" selected>Canada</option>
  <!-- Pre-selected -->
  <option value="uk">United Kingdom</option>
  <option value="au">Australia</option>
</select>

<!-- Multiple select -->
<label for="skills">Skills (hold Ctrl to select multiple):</label>
<select id="skills" name="skills" multiple size="4">
  <option value="html">HTML</option>
  <option value="css">CSS</option>
  <option value="js">JavaScript</option>
  <option value="php">PHP</option>
  <option value="python">Python</option>
</select>

<!-- Option groups -->
<label for="food">Favorite Food:</label>
<select id="food" name="food">
  <optgroup label="Italian">
    <option value="pizza">Pizza</option>
    <option value="pasta">Pasta</option>
    <option value="risotto">Risotto</option>
  </optgroup>
  <optgroup label="Asian">
    <option value="sushi">Sushi</option>
    <option value="ramen">Ramen</option>
    <option value="curry">Curry</option>
  </optgroup>
</select>

```

```

        </optgroup>
    </select>

    <!-- Button elements -->
    <button type="submit">Submit Form</button>
    <button type="reset">Clear Form</button>
    <button type="button" onclick="validateForm()">Validate</button>

    <!-- Custom button with content -->
    <button type="submit">
        
        Submit Application
    </button>

```

Form Validation and Attributes

HTML5 Validation Attributes

```

<!-- Required fields -->
<input type="text" name="name" required>
<input type="email" name="email" required>

<!-- Length validation -->
<input type="text" name="username"
    minlength="3" maxlength="20"
    placeholder="3-20 characters">

<!-- Number validation -->
<input type="number" name="age"
    min="18" max="120"
    step="1">

<!-- Pattern validation (regex) -->
<input type="text" name="phone"
    pattern="[0-9]{3}-[0-9]{3}-[0-9]{4}"
    placeholder="123-456-7890"
    title="Phone format: 123-456-7890">

<!-- Multiple emails -->
<input type="email" name="emails" multiple
    placeholder="email1@example.com, email2@example.com">

<!-- File validation -->
<input type="file" name="resume"
    accept=".pdf,.doc,.docx"
    required>

<!-- Custom validation message -->
<input type="email" name="email" required
    title="Please enter a valid email address">

```

Form Control Attributes

```

<!-- Autofocus -->
<input type="text" name="search" autofocus>

```

```

<!-- Autocomplete -->
<input type="text" name="firstName" autocomplete="given-name">
<input type="text" name="lastName" autocomplete="family-name">
<input type="email" name="email" autocomplete="email">
<input type="tel" name="phone" autocomplete="tel">

<!-- Disabled and readonly -->
<input type="text" name="id" value="12345" readonly>
<input type="text" name="locked" value="Cannot edit" disabled>

<!-- Placeholder text -->
<input type="text" name="search" placeholder="Search products...">
<textarea name="message" placeholder="Enter your message here...">
</textarea>

<!-- Form association -->
<input type="text" name="external" form="mainForm"> <!-- Outside
form -->
<form id="mainForm" action="submit.php" method="post">
  <!-- This input is associated with the form even though it's
outside -->
</form>

<!-- List attribute with datalist -->
<input type="text" name="browser" list="browsers">
<datalist id="browsers">
  <option value="Chrome">
  <option value="Firefox">
  <option value="Safari">
  <option value="Edge">
</datalist>

```

Advanced Form Features

Form Organization

```

<form action="register.php" method="post">
  <!-- Group related fields -->
  <fieldset>
    <legend>Account Information</legend>

    <div class="form-group">
      <label for="username">Username:</label>
      <input type="text" id="username" name="username" required>
    </div>

    <div class="form-group">
      <label for="password">Password:</label>
      <input type="password" id="password" name="password" required>
    </div>
  </fieldset>

  <fieldset>
    <legend>Personal Details</legend>

    <div class="form-group">
      <label for="fullName">Full Name:</label>

```

```

        <input type="text" id="fullName" name="fullName" required>
    </div>

    <div class="form-group">
        <label for="birthdate">Date of Birth:</label>
        <input type="date" id="birthdate" name="birthdate">
    </div>
</fieldset>

<fieldset>
    <legend>Preferences</legend>

    <div class="checkbox-group">
        <input type="checkbox" id="newsletter" name="preferences"
value="newsletter">
        <label for="newsletter">Email newsletter</label>
    </div>

    <div class="checkbox-group">
        <input type="checkbox" id="sms" name="preferences"
value="sms">
        <label for="sms">SMS notifications</label>
    </div>
</fieldset>

<div class="form-actions">
    <button type="submit">Create Account</button>
    <button type="reset">Clear Form</button>
</div>
</form>

```

Complex Form Example

```

<form action="job-application.php" method="post"
enctype="multipart/form-data">
    <h1>Job Application Form</h1>

    <!-- Personal Information -->
    <fieldset>
        <legend>Personal Information</legend>

        <div class="form-row">
            <div class="form-col">
                <label for="firstName">First Name *</label>
                <input type="text" id="firstName" name="firstName" required>
            </div>

            <div class="form-col">
                <label for="lastName">Last Name *</label>
                <input type="text" id="lastName" name="lastName" required>
            </div>
        </div>

        <div class="form-group">
            <label for="email">Email Address *</label>
            <input type="email" id="email" name="email" required
                autocomplete="email">
        </div>

        <div class="form-group">

```

```

        <label for="phone">Phone Number</label>
        <input type="tel" id="phone" name="phone"
            pattern="[0-9]{3}-[0-9]{3}-[0-9]{4}"
            placeholder="123-456-7890">
    </div>

    <div class="form-group">
        <label for="address">Address</label>
        <textarea id="address" name="address" rows="3"></textarea>
    </div>
</fieldset>

<!-- Job Information -->
<fieldset>
    <legend>Job Information</legend>

    <div class="form-group">
        <label for="position">Position Applied For *</label>
        <select id="position" name="position" required>
            <option value="">Select a position</option>
            <optgroup label="Technical">
                <option value="developer">Software Developer</option>
                <option value="designer">UI/UX Designer</option>
                <option value="analyst">Systems Analyst</option>
            </optgroup>
            <optgroup label="Management">
                <option value="manager">Project Manager</option>
                <option value="director">Technical Director</option>
            </optgroup>
        </select>
    </div>

    <div class="form-group">
        <label for="experience">Years of Experience</label>
        <input type="number" id="experience" name="experience"
            min="0" max="50" value="0">
    </div>

    <div class="form-group">
        <label for="salary">Expected Salary Range</label>
        <input type="range" id="salary" name="salary"
            min="30000" max="200000" step="5000" value="60000">
        <output for="salary">$60,000</output>
    </div>

    <div class="form-group">
        <label for="startDate">Available Start Date</label>
        <input type="date" id="startDate" name="startDate">
    </div>
</fieldset>

<!-- Skills and Preferences -->
<fieldset>
    <legend>Skills & Preferences</legend>

    <div class="form-group">
        <label>Programming Languages (select all that apply):</label>
        <div class="checkbox-group">
            <input type="checkbox" id="html" name="skills[]"
value="html">

```

```

        <label for="html">HTML/CSS</label>

        <input type="checkbox" id="javascript" name="skills[]"
value="javascript">
        <label for="javascript">JavaScript</label>

        <input type="checkbox" id="python" name="skills[]"
value="python">
        <label for="python">Python</label>

        <input type="checkbox" id="java" name="skills[]"
value="java">
        <label for="java">Java</label>
    </div>
</div>

<div class="form-group">
    <label>Work Preference:</label>
    <div class="radio-group">
        <input type="radio" id="remote" name="workType"
value="remote">
        <label for="remote">Remote</label>

        <input type="radio" id="office" name="workType"
value="office">
        <label for="office">Office</label>

        <input type="radio" id="hybrid" name="workType"
value="hybrid">
        <label for="hybrid">Hybrid</label>
    </div>
</div>
</fieldset>

<!-- Documents -->
<fieldset>
    <legend>Documents</legend>

    <div class="form-group">
        <label for="resume">Resume/CV *</label>
        <input type="file" id="resume" name="resume"
            accept=".pdf,.doc,.docx" required>
    </div>

    <div class="form-group">
        <label for="portfolio">Portfolio (optional)</label>
        <input type="file" id="portfolio" name="portfolio"
            accept=".pdf,.zip" multiple>
    </div>
</fieldset>

<!-- Additional Information -->
<fieldset>
    <legend>Additional Information</legend>

    <div class="form-group">
        <label for="coverLetter">Cover Letter</label>
        <textarea id="coverLetter" name="coverLetter" rows="6"
            placeholder="Tell us why you're interested in this
position..."></textarea>

```

```

        </div>

        <div class="form-group">
            <label for="referral">How did you hear about us?</label>
            <input type="text" id="referral" name="referral"
list="referralSources">
            <datalist id="referralSources">
                <option value="Job board">
                <option value="Company website">
                <option value="Social media">
                <option value="Referral">
                <option value="Recruiter">
            </datalist>
        </div>
    </fieldset>

    <!-- Terms and Submission -->
    <fieldset>
        <legend>Terms & Conditions</legend>

        <div class="checkbox-group">
            <input type="checkbox" id="terms" name="terms"
value="accepted" required>
            <label for="terms">I agree to the terms and conditions *
        </label>
        </div>

        <div class="checkbox-group">
            <input type="checkbox" id="privacy" name="privacy"
value="accepted" required>
            <label for="privacy">I agree to the privacy policy *</label>
        </div>
    </fieldset>

    <!-- Form Actions -->
    <div class="form-actions">
        <button type="submit">Submit Application</button>
        <button type="reset" onclick="return confirm('Clear all form
data?')">
            Clear Form
        </button>
        <button type="button" onclick="saveAsDraft()">Save as
Draft</button>
    </div>
</form>

```

Form Processing Methods

```

<!-- GET method - data in URL -->
<form action="search.php" method="get">
    <input type="text" name="query" placeholder="Search...">
    <button type="submit">Search</button>
</form>

<!-- Results in URL: search.php?query=searchterm -->

<!-- POST method - data in request body -->
<form action="login.php" method="post">
    <input type="email" name="email" required>
    <input type="password" name="password" required>
    <button type="submit">Login</button>

```

```
</form>

<!-- File upload requires POST and multipart encoding -->
<form action="upload.php" method="post" enctype="multipart/form-
data">
    <input type="file" name="document" accept=".pdf" required>
    <button type="submit">Upload</button>
</form>
```

This comprehensive reference covers all aspects of HTML tables and forms from your PDF, including structure, attributes, styling, validation, and advanced features.